

ERNESTO MIRÓ PERAZA

Senior Software Engineer | Full-Stack Development | Cloud Architecture | Technical Leadership

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PROFESSIONAL SUMMARY

Computer Science Engineer and Senior Software Engineer with 9 years of experience designing, building, and evolving enterprise software. Combines hands-on full-stack development with cloud architecture, technical leadership, and delivery coordination across multi-tenant SaaS, distributed systems, identity, security, and complex integrations. Experienced in modernizing existing platforms and delivering new products across .NET, Angular, GCP, and Azure, with AI-assisted engineering practices supporting planning, documentation, review, and delivery.

SELECTED TECHNICAL SKILLS

Backend & APIs: C#/.NET (.NET Framework 4.7.2–.NET 10), ASP.NET Core, Node.js, TypeScript, Python, REST, gRPC, GraphQL, Apollo Server, SSE.

Architecture: Microservices, event-driven architecture, multi-tenant SaaS, API gateways, state machines, sagas, callbacks, fan-out, webhooks, retry queues, idempotency, and long-running operations.

Frontend: Angular, Angular Material, RxJS, Signals, reactive forms, standalone components, typed GraphQL, PWA, MSAL.js, and WebRTC viewer integration.

Cloud & Infrastructure: GCP (Cloud Run, Cloud Functions, Pub/Sub, Cloud Tasks, Firestore, BigQuery, Cloud KMS, Secret Manager, Compute Engine, Cloud Build); Azure (Azure Virtual Desktop, ARM, Entra ID, Microsoft Graph, Managed Identity); Terraform, Docker, and CI/CD.

Identity, Security & Data: OAuth 2.0, OIDC, SAML 2.0, JWT, JWE, JWKS, SSO, RBAC, envelope encryption, HMAC blind indexing, rate limiting, SQL Server, PostgreSQL, Firestore, BigQuery, Redis, Entity Framework Core, and LINQ.

Engineering & AI: SOLID, Clean Code, design patterns, dependency injection, refactoring, code review, structured logging, feature flags, Git, Scrum, Kanban, Azure DevOps, Jira, Claude Code, Codex, MCP, skills, subagents, and backlog-to-PR workflows.

Integrations: Clever, ClassLink/OneRoster, Google Workspace, Google Admin Directory, Google Classroom API, and Microsoft Graph.

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer & Technical Lead

Private Automotive-Sector Client | Mexico | Remote | 2026

- Led a three-person team while owning full-stack development, architecture, infrastructure preparation, delivery coordination, and client communication.
- Designed a decoupled enterprise CRM architecture and led development with Angular, TypeScript, .NET, ASP.NET Core, Entity Framework Core, and SQL Server.
- Delivered multi-company and multi-branch workflows for customers, vehicles, services, quotations, automotive glass, appointments, users, and configuration.
- Built scheduling capabilities with working hours, holidays, per-slot capacity, availability rules, and conflict validation.
- Implemented JWT authentication in HttpOnly cookies, authorization, session control, validation, centralized error handling, responsive administration, and real-time SSE communication.
- Managed planning, task assignment, progress reviews, blocker resolution, and technical preparation of deployment infrastructure.

Senior Full-Stack Software Engineer

Itopia | United States | Remote | 2019–2026

Technical Leadership & Architecture

- Provided technical leadership for a small frontend, backend, and QA team through planning, estimation, implementation guidance, PR review, and delivery coordination.

- Architected IGW (Itopia Gateway), defining domains, integrations, conventions, and technical criteria for its evolution.
- Standardized structured logging, dependency injection, secret management, and code conventions across more than five repositories.
- Designed AI-assisted engineering workflows using skills, subagents, MCP servers, and documentation as context for planning and PR creation.

PROFESSIONAL EXPERIENCE — CONTINUED

Senior Full-Stack Software Engineer

Itopia | United States | Remote | 2019–2026

Next-Generation Multi-Tenant SaaS Platform | 2025–2026

- Designed and implemented microservices for a multi-tenant SaaS platform serving school districts, with a single-entry gateway, REST APIs, OAuth 2.0 client credentials, scoped authorization, rate limiting, Terraform, and Cloud Build.
- Built session, organization, catalog, and internal event-delivery domains using .NET, gRPC, and service-to-service OIDC authentication.
- Automated Azure tenant onboarding with state machines, Entra ID consent flows, callbacks, event fan-out, and scheduled retries.
- Designed Azure Virtual Desktop capacity and cost controls using ARM, batch startup, disk SKU adjustments, capacity buffers, and controlled pool growth.
- Implemented long-running operations, polling, ghost-session cleanup, feature flags, multi-layer idempotency, strict token validation, and HMAC indexes for sensitive data.

EdTech VDI Platform | 2019–2025

- Built and evolved approximately 20 C#/.NET and Node.js/TypeScript services deployed on Cloud Run and Cloud Functions.
- Developed rostering and identity integrations with Clever, ClassLink/OneRoster, Google Workspace, Google Classroom, Microsoft Graph, Entra ID, OAuth 2.0, and SAML 2.0.
- Implemented usage metering in partitioned BigQuery, VM orchestration in Compute Engine, and event-driven processing with Pub/Sub, retry queues, callbacks, and a real-time SSE gateway.
- Built Angular portals for administration, end users, and support, with typed GraphQL, PWA capabilities, and remote-desktop viewer integration.
- Developed a .NET Windows agent for VDI virtual machines and implemented envelope encryption, secret rotation, Terraform infrastructure, and Cloud Build pipelines.
- Diagnosed and evolved the WebRTC/GStreamer streaming stack while contributing to WorkAnyWhere, Cloud Compliance Manager, CloudApps, and ItopiaOne.

Project Specialist

XETID | Cuba | 2017–2019

- Developed and maintained enterprise backend and frontend capabilities in the Entity Management Division and contributed to the Distra ARP project.
- Implemented features with PostgreSQL, Zend Framework, Ext JS, and Bootstrap while analyzing requirements and resolving functional and technical needs within a multidisciplinary team.

Software Engineer — Academic Project

University of Information Sciences (UCI) | Cuba | 2016–2017

- Developed SBRSPPLIT, a final-degree system predicting compatibility between students and courses, using Python, PyQt5, PostgreSQL, Excel-processing libraries, and PyInstaller.

EDUCATION & LANGUAGES

Computer Science Engineering

University of Information Sciences (UCI) | Cuba | 2012–2017

Languages: Spanish — native. English — professional working proficiency in conversations and meetings, with advanced reading, writing, and technical vocabulary.